

Data_Scientist — Step 3

Learn Data Analysis

Detailed Learning Notes • CareerCompass

1. Learning Objective

Learn a disciplined workflow for turning raw data into reliable, interpretable findings.

2. Core Concepts — Detailed Breakdown

Data profiling

Inspect schema, ranges, unique values, missingness, duplicates and suspicious categories before analysis.

Cleaning

Define rules for missing values, invalid records, duplicates and outliers. Keep a record of what was changed and why.

Exploratory analysis

Use grouping, aggregation, distributions and relationships to generate hypotheses. EDA is not a license to cherry-pick interesting patterns.

Feature engineering

Create meaningful variables such as ratios, rolling metrics, time components and grouped categories while preserving the analytical question.

Insight generation

Every finding should connect a number or visual pattern to a question. Include uncertainty, alternative explanations and limitations.

3. Recommended Learning Workflow

- Profile first.
- Create a data dictionary.
- Clean with explicit rules.
- Explore one question at a time.
- Validate surprising results with an independent query or calculation.
- Write conclusions separately from observations.

5. Hands-on Project / Practice

Perform a full EDA on a public dataset. Include a data-quality report, cleaning log, at least eight analytical questions, grouped summaries, distribution and relationship plots, and a final executive-style summary.

6. Common Mistakes & Pitfalls

- Deleting outliers automatically.
- Filling all missing values with zero.
- Making dozens of plots without a question.

- Changing data until a desired result appears.
- Ignoring duplicate records or unit mismatches.

7. Completion Checklist

- I can profile a dataset.
- I can document cleaning decisions.
- I can perform grouped analysis.
- I can validate an insight with another method.
- I can write a concise analytical conclusion.

8. Interview / Assessment Questions

- What should you inspect before modeling?
- How do you decide whether to remove an outlier?
- Why can missing values be informative?
- What is EDA?
- How do you validate a surprising insight?

9. Recommended References

pandas User Guide: https://pandas.pydata.org/docs/user_guide/

Data Analysis with Python: <https://www.youtube.com/watch?v=GPVsHOIRBBI>

Python for Data Science: <https://www.youtube.com/watch?v=CMEWVn1uZpQ>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.